

THE SOUND- AND LIGHT- PRODUCING ORGANS

Sound-producing Organs

Sounds of different kinds and intensities are produced by a number of species in all the main orders of insects; for general reviews see Alexander (1967), Dumortier (1963*a, b* and *c*), Frings and Frings (1958, 1960), Haskell (1961), Michelsen and Nocke (1974) and Bennet-Clark (1975). In some cases, the sound-producing organs are similar in the two sexes (as in many Coleoptera) but often they are confined to, or more strongly developed in, the male (e.g. most Orthoptera and Cicadidae). The biological significance of the sounds which are produced is not always clear but in some species they facilitate mating by attracting the sexes or stimulating the female and in other instances they may express sexual rivalry between males, subserve species-recognition (thus helping to keep members of the same species together), communicate warnings of danger or have a defensive function. A single species may make several sounds, each with its own function.

The methods by which sounds are produced may be classified under the following main headings.

- (a) By tapping part of the body against an external object.
- (b) By friction of one part of the body against another.
- (c) By vibration of the wings or thoracic wall.
- (d) By vibration of a special membrane through muscular action.
- (e) By emission of air.

A. Sounds Produced by Tapping Part of the Body against an External Object – The best-known example of this type of sound-production occurs in the Death-watch Beetle, *Xestobium rufovillosum*, and allied Anobiidae where what is thought to be a sexual call is made by tapping the head against the walls of the burrow (Darwin, 1909). Some male Plecoptera rap the end of the abdomen against the substrate (Rupprecht, 1968) and the Psocopteran *Trogium pulsatorium* (the 'Lesser Death-watch') also makes a sound in this way (Pearman, 1928). Again, the pupae of a few

Hesperiidae and Lycaenidae produce what is probably a defensive sound by knocking the body against the walls of their cells or the substrate (Hinton, 1948). Finally, the soldiers of many species of termites (e.g. *Bellicositermes*) respond to slight mechanical disturbances of their nest by hammering their heads against the nest-structure, sometimes in rhythmic unison (Emerson, 1930). The resulting vibrations are transmitted through the substrate and probably act as a warning signal (Howse, 1964).

B. Sounds Produced by Friction of one Part of the Body against Another – Most of the sounds emitted by insects are produced by this method, the parts concerned being known as stridulating organs. Practically every external part of the body which is subjected to friction on an adjoining part has given rise to a stridulating organ in one or other insect.

Stridulating organs are possessed by representatives of several orders of insects, particularly the Orthoptera, Coleoptera and Hemiptera, but it is in the first mentioned order that they are best known (e.g. Faber, 1953; Haskell, 1957–60; Jacobs, 1953, and others cited below). In many of the Acridoidea, Tettigonioidae and Grylloidea the males are capable of vigorous stridulation; outside these three groups fewer other Orthoptera stridulate and the faculty is less often present in the females (Ragge, 1955; Spooner, 1968). Among the Acrididae the sounds are often produced by a row of pegs on the inner side of each hind femur being worked against the outer surface of each tegmen (Fig. 89 and see Evans, 1952; Loher, 1957; Weih, 1951; Otte, 1970).



FIG. 89 Hind femur of an Acridid

a, row of pegs, three of which are shown greatly enlarged.

In some others (Oedipodinae) a row of prominences on a secondary tegminal vein is scraped by a ridge on the inner side of each hind femur. In the Tettigoniidae and Gryllidae the sound is produced when a row of teeth on the tegminal vein Cu_2 (the file) is scraped by a sclerotized part of the margin of the opposite tegmen, thus throwing into vibration certain areas of the tegmen (Stärk, 1958; Bailey, 1970; Bailey and Broughton, 1970; Nielsen and Dreisig, 1970; Bennet-Clark, 1970, and Nocke, 1971). Yet another type of stridulatory apparatus occurs in the Gryllacrididae where ridges on the side of the 2nd or 3rd abdominal segments are rubbed by the posterior femora and several other mechanisms peculiar to individual species of Orthoptera have been described (e.g. by Kevan, 1955). The sounds produced are often very characteristic of the species and each form may produce several types of sound including a normal song and a number of epigamic songs (Faber,

1929–36; Jacobs, 1950). Analysis of Orthopteran stridulation by electrical methods (Pierce, 1948 and much recent work) reveals the complex character of the sound, e.g. in *Gryllus assimilis* the normal song consists of a series of chirps at intervals of about $\frac{1}{3}$ of a second, each chirp comprising 2–6 very short pulses during which sound-waves with a frequency of about 5 kHz undergo amplitude modulation. In *Locusta migratoria* the greatest intensity of sound is emitted at about 12.5 kHz (Evans, *l.c.*) while artificial stimulation of the tegmen of *Homorocoryphus* produces an almost pure tone of 15 kHz (Bailey, *l.c.*). The songs of Orthoptera are generally characteristic of the species and sometimes allow morphologically very similar sibling species to be distinguished (e.g. Alexander and Thomas, 1959; Madsen *et al.*, 1970; Walker, 1964, 1969*a, b*). In a few well-studied cases the epigamic songs have been shown to ensure conspecific matings and thus maintain reproductive isolation in sympatric species (e.g. Perdeck, 1957).

Among Coleoptera, stridulatory organs have been reported from widely different parts of the body in a variety of species, both from larvae (e.g. *Geotrupes*) and adults (Gahan, 1900; Dudich, 1920–21). One part of the exoskeleton forms a file-like area (*pars stridulans*) which is rubbed by an adjacent region (*plectrum*) when the two undergo relative movement. The organs are generally inconspicuous and equally developed in both sexes; in a few species they are confined to one sex, the female in *Phanopate* but more usually the male (e.g. *Xenoderus*, *Cryptorrhynchus lapathi*). Stridulation sometimes has a sexual significance in beetles. The weevil *Rhynchaenus fagi* stridulates during courtship (Claridge, 1968) and in several species of the Scolytid *Ips* the females stridulate when entering the entrance tunnels made by the males; if they are stopped experimentally from stridulating the male prevents them from entering (Barr, 1969).

Among Hemiptera there is similarly a great diversity of stridulatory organs of this type, especially among Heteroptera (Leston, 1956), where both sexes or the male alone can produce sounds. The peculiar strigil of male Corixidae is apparently used in some unknown way to produce a sound in *Micronecta*, but in those other male Corixids which stridulate the noise is made by the passage of an area of spines on the femur over the angular side-margins of the face (von Mitis, 1935; Finke, 1968; Jansson, 1973).

Several Lepidoptera are known to be capable of stridulation (Hannemann, 1956). They usually produce a hissing or rustling sound and four mechanisms may be noted: (a) friction between the fore and hind wings, as in *Cidaria dotata* males (Hampson, 1894); (b) friction between legs and wings, as in some male Agaristinae (Hampson, 1892) and fertilized females of *Parnassius mnemosyne* (Jobling, 1936); (c) friction between adjacent parts of the thoracic wall (e.g. both sexes of the Arctiid *Rhodogastria*, where the sound is accompanied by the emission of froth through reflex bleeding and is probably defensive (Carpenter, 1938)); (d) friction between part of the 3rd abdominal sternum and an infolding of the adjacent sternopleural region, as in some male Lymantriids.

In the Hymenoptera stridulating organs are common among certain ants and vary in structure in different species and in the castes of the same species (Fig. 90). The organ commonly consists of a file on the first gastral tergum which is scraped by the posterior edge of the preceding tergum. Workers of *Atta cephalotes* produce sounds which are perceived by other workers only when conducted through the soil (with maximum intensity in the frequency range 1–3 kHz). In this way workers that have been accidentally buried can attract others that dig them out (Markl, 1967, 1968).



FIG. 90 Stridulating organ of *Myrmica rubra* (= *laevinodis*) in median section
p, edge of postpetiole forming a 'scraper'; *s*, stridulatory surface on first gastral segment; *i*, intersegmental membrane. After Janet.

C. Sounds Produced by Vibration of the Wings or Thoracic Wall –

Many insects make a humming or buzzing sound during flight, the frequency of the emitted note appearing to some observers to be the same as that of the wing-beat. Sotavalta (1947) gives numerous examples, but in a more detailed physical analysis of the flight-tone of *Drosophila funebris* Williams and Galambos (1950) found that though each cycle of wing-movement corresponds to one of sound, there are, in addition to the fundamental frequency, prominent harmonics which result in wide departures from a simple sinusoidal type of vibration (see also Sotavalta, 1963). In some Diptera and Hymenoptera there is, in addition to the main wing-beat tone and its harmonics, an independent sound of higher frequency occurring on each half-cycle of the wing-beat and perhaps due to vibration of the thoracic exoskeleton (Esch and Wilson, 1967). This sound can also be produced when the wings are folded and in *Apis* a colony about to swarm will do so only after its members have received such a sound-signal from certain bees (Esch, 1967*b*). *Melipona* workers returning after a successful foraging flight can convey information on the site of a suitable food source by producing sounds, again with closed wings, at frequencies of 300–800 Hz, as compared with the normal flight tone of 200 Hz (Esch, 1967*a*). The males of various species of *Drosophila* emit characteristic courtship songs by special wing vibrations. These consist of regularly repeated pulses of sound and allow the females to recognize conspecific males (Bennet-Clark and Ewing, 1968; Ewing and Bennet-Clark, 1968).

D. Sounds Produced by the Vibration of a Special Membrane Exerted by Muscular Action – Sound-producing organs of this type are characteristic of the Auchenorrhynchan Homoptera and among the Cicadidae there is found one of the most complex kinds of sound-producing organs known. These structures are met with in the males, the females being either silent or only possessing rudiments of the apparatus. The great volume of sound emitted by the cicadas marks them out as being the noisiest representatives of the Insecta.

The apparatus comprises a pair of shell-like drums or tymbals situated at the base of the abdomen and vibrating by the action of powerful muscles

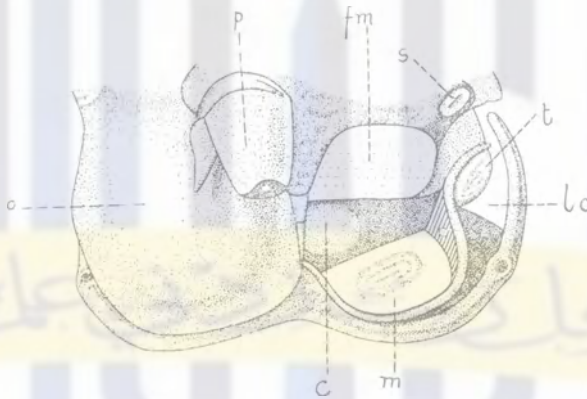


FIG. 91 Sound-producing apparatus of a cicada with the operculum of one side removed

c, ventral cavity; *fm*, folded membrane; *lc*, lateral cavity; *m*, tympanal organ; *o*, operculum, that of the other side removed; *p*, base of leg; *s*, spiracle; *t*, tymbal. After Carlet, *Ann. Sci. nat.*, Paris, 1887.

(Pringle, 1954, 1957). In *Magicicada septendecim* the true sound organs are freely exposed, but in many other cicadas the drums are covered by overlapping *opercula*, a pair of large plates which are backward extension of the metathoracic epimera and situated on the ventral side of the body, where they overlap the base of the abdomen (Fig. 91). On removing an operculum a pair of cavities containing the external parts of the sound-producing apparatus is disclosed. The larger of these cavities is ventral, and the smaller is lateral in position. Their walls contain three specialized areas of membrane which are known respectively as the *tymbal*, the *folded membrane* and the *mirror*. The *tymbal* is a crisp, plaited membrane surrounded by a sclerotized ring; it forms part of the inner wall of the lateral cavity, and is somewhat shell-like in appearance with its convex surface bulging outwards. The *folded membrane* is in the anterior wall of the ventral cavity, while the *mirror* is a tense, transparent membrane in the posterior wall of that cavity. In close association with the whole apparatus there is an extensive air-chamber which

opens to the exterior by the third pair of spiracles. The sound is produced by the rapid in and out movement of the tymbal, which is brought about by a powerful muscle arising from the 2nd abdominal sternum and attached to the inner face of the tymbal. When the muscle contracts the tymbal is pulled inwards; on relaxation of the muscle the tymbal regains its former position by virtue of the elasticity of its cuticular ring. This method of sound production has been compared to the pushing in and out of the bottom of a tin vessel, which makes a cracking sound.

Electrophysiological analysis shows the patterns of sound produced by cicadas to be relatively complex (Pringle, *op. cit.*, Aidley, 1969, Young, 1972). The tymbal muscles may contract simultaneously or alternately and they may be synchronous or asynchronous with respect to the nerve-impulses reaching them. The sound waves, at frequencies from 850 to several thousand Hz, are amplitude-modulated to produce pulses repeated at the rate of 40 to over 1000 per second. In *Abricta* and *Magicicada* the strongly-developed ribs of the tymbal buckle separately during the 'in' movement, so that each muscle contraction produces 7-9 sound pulses. Among Australian cicadas the enlarged abdomen of *Cystosoma saundersi* acts as a resonating chamber, while in *Arunta perulata* the enlarged opercula impart strong directional properties to the sound. Cicada songs differ appreciably from one species to another (see also Alexander and Moore, 1958) and their function is apparently to bring the individuals together and to attract females to males (Myers, 1929), though distress songs have also been described.

Ossiannilsson (1949) has found that males in many other Auchenorrhynchan families (Delphacidae, Cixiidae, Cercopidae, Membracidae, Cicadellidae) have small stridulatory organs essentially similar to those of cicadas and more weakly developed organs even occur in some female Cercopids and Cicadellids. The sounds produced by these insects are always of low intensity and not always audible to the unaided ear, though several types of song (courtship, rivalry, distress, etc.) are said to be distinguishable. See also Strübing (1962, 1965) and Claridge and Howse (1968).

A pair of sound-producing tymbals also occurs in the Arctiid moth *Melese laodamia*, where each organ is a swollen, air-filled structure formed externally from the metathoracic katapisternum, with 15-20 striae on its outer face. It can be compressed by the basalar muscles, when the striae act as an array of 'microtymbals', each producing a brief pulse of sound. The overall result is a pattern of ultrasonic sounds generated between 30 and 90 kHz with a pulse repetition frequency of 1200 per second (Blest, Collett and Pye, 1963). Similar organs occur in some other Arctiids which are also known to be distasteful and to exhibit a protective display when stimulated (Blest, 1964).

E. Sounds Associated with the Emission of Air – Older authors considered that the humming of bees and flies was due to the vibration of

lamellae just inside the thoracic spiracles but this idea has now been abandoned. More recently, the 'piping' of newly-emerged queens of *Apis mellifera* has been ascribed to the emission of air from the spiracles (Woods, 1956) though it is perhaps more likely to be due to thoracic vibration of the kind described above in section C (Simpson, 1964). The cry of the Death's Head Hawkmoth, *Acherontia atropos*, has puzzled many investigators but is probably due, as Prell (1920) first showed, to the emission of air from the pharynx near the base of the proboscis. Two sounds are emitted: a low note accompanied by vibration of the epipharynx and a high-pitched whistle that occurs when air passes out freely (Busnel and Dumortier, 1959).

The Light-Producing or Photogenic Organs

Certain insects are self-luminous owing to the possession of special photogenic organs; others owe their luminous properties to the presence of light-producing bacteria, or through having ingested luminous food (McElroy, 1964). True luminous insects are almost confined to the order Coleoptera and more particularly to various genera of Lampyridae and Cantharidae, notably *Lampyris*, *Luciola*, *Phosphoenus*, *Photuris*, *Photinus*, *Phengodes* as well as the Elaterid genera *Pyrophorus* (the 'cucujos') and *Photophorus*. Outside the Coleoptera, the Mycetophilids *Arachnocampa luminosa* and *Ceroplatus testaceus* have photogenic powers (Stammer, 1932; Gatenby, 1960). Their luminous larvae spin webs in which they catch prey, presumably attracted by the light. In *Arachnocampa* the light of older female pupae and adult females attracts the males (which are also luminous, though their light does not seem to be used epigamically); see Richards, A. M. (1960). Other examples of luminescence from the Collembola, Fulgoridae and a few other groups are not sufficiently understood (Harvey, 1957).

In the Lampyridae the luminosity is known in some species to extend to all the developmental stages, and is a character of their cytoplasm. In the egg the luminous substance is diffused, but in the postembryonic phases it is localized in the photogenic organs. The latter structures vary greatly in size, shape and position in different species and according to the sex and developmental stage. In *Phengodes* and *Phrixothrix*, for example, there are 11 or 12 segmentally arranged pairs of organs while in many species (e.g. *Lampyris noctiluca* females) they are found principally on the 6th and 7th abdominal sterna and in *Pyrophorus* they occur on the pronotum and anterior abdominal sterna. They may be equally developed in both sexes, as in *Luciola* and the Elateridae, or the females, which are sometimes apterous and larviform, may be the sole or principal luminous sex.

According to the species concerned, the light may be emitted as a continuous glow, an intermittent or pulsating glow of variable intensity or periodic short flashes whose characteristics are influenced by environmental factors e.g. Seliger *et al.* (1964). The spectacular synchronous flashing of Lampyrids (Buck and Buck, 1968; Baldaccini *et al.*, 1969) is not fully explained. The

light occupies wavelengths over the range 5000 to 6500 Å and is usually yellowish-green though in *Pyrophorus* the abdominal organs emit a reddish light in flight and the thoracic ones a green light when the insect is not flying. The light is usually thought to assist in the attraction of the sexes for mating (e.g. Papi, 1969) and two main systems of communication have been distinguished (Lloyd, 1966, 1971). In some species one sex (usually the female) is sedentary and emits a species-specific light-signal that attracts the opposite sex (e.g. *Lampyris noctiluca*, *Phengodes laticollis*). In others the

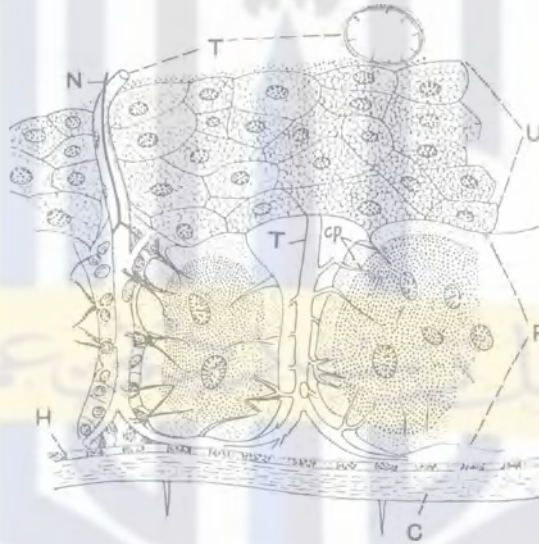


FIG. 92 Luminous organ of *Photinus*

C, cuticle; cp, tracheoles; H, epidermis; N, nerve; P, photocyte layer; T, tracheae; U, reflector layer. After Williams, 1916.

flying male emits a specific signal, the female responds by another signal and the male then flies towards her, as in *Photuris* and *Photinus*. In the synchronously flashing fireflies of S.E. Asia (*Pteroptyx*; Ballantyne and McLean, 1970) aggregations of synchronizing males attract both males and females, after which the females may respond and attract the males.

The organs are located beneath areas of transparent cuticle and their histological structure shows various degrees of complexity from those of *Phengodes*, where the photogenic structures (photocytes) are large independent cells similar to oenocytes, to those where a compact organ (Fig. 92) is composed of masses of photocytes arranged in rosette-like fashion around tracheae from which short trunks run into tracheal end-cells. From the latter arise numerous tracheoles which penetrate between the photogenic cells while the inner surface of the organ is covered by a layer of cells containing

inclusions thought to be urates. These form the so-called reflecting layer, the function of which is uncertain.

The photocytes are packed with granules about $2.5\ \mu\text{m}$ in diameter and the organ contains specialized nerve terminals. The axoplasm of these possess granules which are structurally identical with synaptic vesicles and neurosecretory material and the nerves end in association with the membranes of the photocytes and the tracheolar and tracheal end-cells. For details see Buck (1948), Smith (1963), Peterson and Buck (1968) and Peterson (1970).

As in other luminous animals and plants, the reactions leading to luminescence begin when an enzyme, *luciferase*, in the form of a complex with adenosine triphosphate, combines with reduced *luciferin* to form a further complex which is itself oxidized by molecular oxygen with the emission of light. The oxidized product is very stable but luciferase can be freed from it by pyrophosphate, thus providing material to start the cycle again (McElroy and Seliger, 1963; McElroy, Seliger and White, 1970). The flashing that occurs in luminous organs of complex histological structure is subject to nervous control, perhaps depending ultimately on the release of pyrophosphate in the photocytes (Carlson, 1969).

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